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6.3 Total Least Squares

The problem of minimizing $\| Ax - b \|_2$ where $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$ can be recast as follows:

$$\min_{b+r \,\in\, \mathrm{ran}(A)} \| r \|_2 \ . \tag{6.3.1}$$

In this problem, there is a tacit assumption that the errors are confined to the *vector of observations* b. If error is also present in the *data matrix* A, then it may be more natural to consider the problem

$$\min_{b+r \,\in\, \mathrm{ran}(A+E)} \| \, [\, E \,|\, r \,] \, \|_F \ . \tag{6.3.2}$$

This problem, discussed by Golub and Van Loan (1980), is referred to as the *total least squares* (TLS) problem. If a minimizing $[\, E_0 \,|\, r_0 \,]$ can be found for (6.3.2), then any x satisfying $(A + E_0)x = b + r_0$ is called a TLS solution. However, it should be realized that (6.3.2) may fail to have a solution altogether. For example, if

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad E_\epsilon = \begin{bmatrix} 0 & 0 \\ 0 & \epsilon \\ 0 & \epsilon \end{bmatrix},$$

then for all $\epsilon > 0$, $b \in \mathrm{ran}(A + E_\epsilon)$. However, there is no smallest value of $\| \, [\, E \,,\, r \,] \, \|_F$ for which $b + r \in \mathrm{ran}(A + E)$.

A generalization of (6.3.2) results if we allow multiple right-hand sides and use a weighted Frobenius norm. In particular, if $B \in \mathbb{R}^{m\times k}$ and the matrices

$$D = \mathrm{diag}(d_1, \ldots, d_m),$$

$$T = \mathrm{diag}(t_1, \ldots, t_{n+k})$$

are nonsingular, then we are led to an optimization problem of the form

$$\min_{B+R \,\in\, \mathrm{ran}(A+E)} \| \, D \, [\, E \,|\, R \,] \, T \, \|_F \tag{6.3.3}$$

where $E \in \mathbb{R}^{m\times n}$ and $R \in \mathbb{R}^{m\times k}$. If $[\, E_0 \,|\, R_0 \,]$ solves (6.3.3), then any $X \in \mathbb{R}^{n\times k}$ that satisfies

$$(A + E_0)X \;=\; (B + R_0)$$

is said to be a TLS solution to (6.3.3).

In this section we discuss some of the mathematical properties of the total least squares problem and show how it can be solved using the SVD. For a more detailed introduction, see Van Huffel and Vanderwalle (1991).

6.3.1 Mathematical Background

The following theorem gives conditions for the uniqueness and existence of a TLS solution to the multiple-right-hand-side problem.

Theorem 6.3.1. *Suppose $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{m\times k}$ and that $D = \mathrm{diag}(d_1,\ldots,d_m)$ and $T = \mathrm{diag}(t_1,\ldots,t_{n+k})$ are nonsingular. Assume $m \geq n+k$ and let the SVD of*

$$C \;=\; D[\,A\,|\,B\,]T \;=\; [\;\underset{n}{C_1}\,|\,\underset{k}{C_2}\,]$$

be specified by $U^TCV = \mathrm{diag}(\sigma_1,\ldots,\sigma_{n+k}) = \Sigma$ where U, V, and Σ are partitioned as follows:

$$U \;=\; [\;\underset{n}{U_1}\,|\,\underset{k}{U_2}\,]\;, \qquad V \;=\; \begin{array}{cc} \left[\begin{array}{cc} V_{11} & V_{12} \\ V_{21} & V_{22} \end{array}\right] & \begin{array}{c} n \\ k \end{array} \\ \begin{array}{cc} n & k \end{array} & \end{array}, \qquad \Sigma \;=\; \begin{array}{cc} \left[\begin{array}{cc} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{array}\right] & \begin{array}{c} n \\ k \end{array} \\ \begin{array}{cc} n & k \end{array} & \end{array}.$$

If $\sigma_n(C_1) > \sigma_{n+1}(C)$, then the matrix $[\,E_0\,|\,R_0\,]$ defined by

$$D[\,E_0\,|\,R_0\,]T \;=\; -U_2\Sigma_2[\,V_{12}^T\,|\,V_{22}^T\,] \tag{6.3.4}$$

solves (6.3.3). If $T_1 = \mathrm{diag}(t_1,\ldots,t_n)$ and $T_2 = \mathrm{diag}(t_{n+1},\ldots,t_{n+k})$, then the matrix

$$X_{TLS} \;=\; -T_1V_{12}V_{22}^{-1}T_2^{-1}$$

exists and is the unique TLS solution to $(A+E_0)X = B+R_0$.

Proof. We first establish two results that follow from the assumption $\sigma_n(C_1) > \sigma_{n+1}(C)$. From the equation $CV = U\Sigma$ we have

$$C_1V_{12} + C_2V_{22} \;=\; U_2\Sigma_2.$$

We wish to show that V_{22} is nonsingular. Suppose $V_{22}x = 0$ for some unit 2-norm x. It follows from

$$V_{12}^TV_{12} + V_{22}^TV_{22} = I$$

that $\|\, V_{12}x \,\|_2 = 1$. But then

$$\sigma_{n+1}(C) \;\geq\; \|\, U_2\Sigma_2x \,\|_2 \;=\; \|\, C_1V_{12}x \,\|_2 \;\geq\; \sigma_n(C_1)\,,$$

a contradiction. Thus, the submatrix V_{22} is nonsingular. The second fact concerns the strict separation of $\sigma_n(C)$ and $\sigma_{n+1}(C)$. From Corollary 2.4.5, we have $\sigma_n(C) \geq \sigma_n(C_1)$ and so

$$\sigma_n(C) \;\geq\; \sigma_n(C_1) \;>\; \sigma_{n+1}(C).$$

We are now set to prove the theorem. If $\mathrm{ran}(B+R) \subset \mathrm{ran}(A+E)$, then there is an X (n-by-k) so $(A+E)X = B+R$, i.e.,

$$\{\, D[\,A\,|\,B\,]T + D[\,E\,|\,R\,]T \,\}\,T^{-1}\left[\begin{array}{c} X \\ -I_k \end{array}\right] \;=\; 0\,. \tag{6.3.5}$$

Thus, the rank of the matrix in curly brackets is at most equal to n. By following the argument in the proof of Theorem 2.4.8, it can be shown that

$$\| D[\,E \,|\, R\,]T \|_F^2 \;\geq\; \sum_{i=n+1}^{n+k} \sigma_i(C)^2.$$

Moreover, the lower bound is realized by setting $[\,E \mid R\,] = [\,E_0 \mid R_0\,]$. Using the inequality $\sigma_n(C) > \sigma_{n+1}(C)$, we may infer that $[\,E_0 \mid R_0\,]$ is the unique minimizer.

To identify the TLS solution X_{TLS}, we observe that the nullspace of

$$\{D[\,A \,|\, B\,]T \;+\; D[\,E_0 \,|\, R_0\,]T\} \;=\; U_1\,\Sigma_1\,[\,V_{11}^T \,|\, V_{21}^T\,]$$

is the range of $\begin{bmatrix} V_{12} \\ V_{22} \end{bmatrix}$. Thus, from (6.3.5)

$$T^{-1}\begin{bmatrix} X \\ -I_k \end{bmatrix} \;=\; \begin{bmatrix} V_{12} \\ V_{22} \end{bmatrix} S$$

for some k-by-k matrix S. From the equations $T_1^{-1}X = V_{12}S$ and $-T_2^{-1} = V_{22}S$ we see that $S = -V_{22}^{-1}T_2^{-1}$ and so

$$X \;=\; T_1V_{12}S \;=\; -T_1V_{12}V_{22}^{-1}T_2^{-1} \;=\; X_{\text{TLS}}. \qquad \square$$

Note from the thin CS decomposition (Theorem 2.5.2) that

$$\| X \|_\tau^2 \;=\; \| V_{12}V_{22}^{-1} \|_2^2 \;=\; \frac{1 - \sigma_k(V_{22})^2}{\sigma_k(V_{22})^2}$$

where we define the "τ-norm" on $\mathbb{R}^{n\times k}$ by $\| Z \|_\tau = \| T_1^{-1}ZT_2 \|_2$.

If $\sigma_n(C_1) = \sigma_{n+1}(C)$, then the solution procedure implicit in the above proof is problematic. The TLS problem may have no solution or an infinite number of solutions. See §6.3.4 for suggestions as to how one might proceed.

6.3.2 Solving the Single Right Hand Side Case

We show how to maximize $\sigma_k(V_{22})$ in the important $k = 1$ case. Suppose the singular values of C satisfy $\sigma_{n-p} > \sigma_{n-p+1} = \cdots = \sigma_{n+1}$ and let $V = [\,v_1 \,|\cdots|\, v_{n+1}\,]$ be a column partitioning of V. If $\widetilde{Q}$ is a Householder matrix such that

$$V(:, n+1-p{:}n+1)\widetilde{Q} \;=\; \begin{array}{cc} \begin{bmatrix} W & z \\ 0 & \alpha \end{bmatrix} & \begin{array}{l} n \\ 1 \end{array} \\ \begin{array}{cc} \;\;p & \;\;\;1 \end{array} & \end{array},$$

then the last column of this matrix has the largest $(n+1)$st component of all the vectors in $\mathsf{span}\{v_{n+1-p}, \ldots, v_{n+1}\}$. If $\alpha = 0$, then the TLS problem has no solution. Otherwise

$$x_{\text{TLS}} = -T_1z/(t_{n+1}\alpha).$$

Moreover,

$$\begin{bmatrix} I_{n-1} & 0 \\ 0 & \tilde{Q} \end{bmatrix} U^T(D[\,A\,|\,b\,]T)V \begin{bmatrix} I_{n-p} & 0 \\ 0 & \tilde{Q} \end{bmatrix} = \Sigma$$

and so

$$D[\,E_0\,|\,r_0\,]T \;=\; -D[\,A\,|\,b\,]T \begin{bmatrix} z \\ \alpha \end{bmatrix} [\,z^T\,|\,\alpha\,].$$

Overall, we have the following algorithm:

Algorithm 6.3.1 Given $A \in \mathbb{R}^{m\times n}$ $(m > n)$, $b \in \mathbb{R}^m$, nonsingular $D = \text{diag}(d_1,\ldots,d_m)$, and nonsingular $T = \text{diag}(t_1,\ldots,t_{n+1})$, the following algorithm computes (if possible) a vector $x_{\text{TLS}} \in \mathbb{R}^n$ such that $(A+E_0)x_{\text{TLS}} = (b+r_0)$ and $\|\, D[\,E_0\,|\,r_0\,]T\,\|_F$ is minimal.

Compute the SVD $U^T(D[\,A\,|\,b\,]T)V \;=\; \text{diag}(\sigma_1,\ldots,\sigma_{n+1})$ and save V.

Determine p such that $\sigma_1 \geq \cdots \geq \sigma_{n-p} > \sigma_{n-p+1} = \cdots = \sigma_{n+1}$.

Compute a Householder P such that if $\tilde{V} = VP$, then $\tilde{V}(n+1, n-p+1{:}n) = 0$.

if $\tilde{v}_{n+1,n+1} \neq 0$

  for $i = 1{:}n$

    $x_i = -t_i\tilde{v}_{i,n+1}/(t_{n+1}\tilde{v}_{n+1,n+1})$

  end

  $x_{\text{TLS}} = x$

end

This algorithm requires about $2mn^2 + 12n^3$ flops and most of these are associated with the SVD computation.

6.3.3 A Geometric Interpretation

It can be shown that the TLS solution x_{TLS} minimizes

$$\psi(x) \;=\; \sum_{i=1}^{m} d_i^2 \left(\frac{|a_i^T x - b_i|^2}{x^T T_1^{-2} x + t_{n+1}^{-2}} \right) \tag{6.3.6}$$

where a_i^T is the ith row of A and b_i is the ith component of b. A geometrical interpretation of the TLS problem is made possible by this observation. Indeed,

$$\delta_i \;=\; \frac{|a_i^T x - b_i|^2}{x^T T_1^{-2} x + t_{n+1}^{-2}}$$

is the square of the distance from

$$\begin{bmatrix} a_i \\ b_i \end{bmatrix} \in \mathbb{R}^{n+1}$$

to the nearest point in the subspace

$$P_x \;=\; \left\{ \begin{bmatrix} a \\ b \end{bmatrix} : a \in \mathbb{R}^n,\ b \in \mathbb{R},\ b = x^T a \right\}$$

where the distance in $\mathbb{R}^{n+1}$ is measured by the norm $\| z \| = \| Tz \|_2$. The TLS problem is essentially the problem of *orthogonal regression*, a topic with a long history. See Pearson (1901) and Madansky (1959).

6.3.4 Variations of the Basic TLS Problem

We briefly mention some modified TLS problems that address situations when additional constraints are imposed on the optimizing E and R and the associated TLS solution.

In the *restricted TLS problem*, we are given $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{m\times k}$, $P_1 \in \mathbb{R}^{m\times q}$, and $P_2 \in \mathbb{R}^{n+k\times r}$, and solve

$$\min_{B+R \,\subset\, \mathsf{ran}(A+E)} \| P_1^T [\, E \,|\, R \,] P_2 \|_F \, . \tag{6.3.7}$$

We assume that $q \le m$ and $r \le n+k$. An important application arises if some of the columns of A are error-free. For example, if the first s columns of A are error-free, then it makes sense to force the optimizing E to satisfy $E(:, 1{:}s) = 0$. This goal is achieved by setting $P_1 = I_m$ and $P_2 = I_{m+k}(:, s+1{:}n+k)$ in the restricted TLS problem.

If a particular TLS problem has no solution, then it is referred to as a *nongeneric TLS problem*. By adding a constraint it is possible to produce a meaningful solution. For example, let $U^T[\, A \mid b \,]V = \Sigma$ be the SVD and let p be the largest index so $V(n+1, p) \neq 0$. It can be shown that the problem

$$\min_{\substack{(A+E)x=b+r \\ [\, E \,|\, r \,]V(:,p+1:n+1)=0}} \| [\, E \,|\, r \,] \|_F \tag{6.3.8}$$

has a solution $[\, E_0 \mid r_0 \,]$ and the nongeneric TLS solution satisfies $(A+E_0)x + b + r_0$. See Van Huffel (1992).

In the *regularized TLS problem* additional constraints are imposed to ensure that the solution x is properly constrained/smoothed:

$$\min_{\substack{(A+E)x=b+r \\ \|Lx\|_2 \le \delta}} \| [\, E \,|\, r \,] \|_F \, . \tag{6.3.9}$$

The matrix $L \in \mathbb{R}^{n\times n}$ could be the identity or a discretized second-derivative operator. The regularized TLS problem leads to a Lagrange multiplier system of the form

$$(A^TA + \lambda_1 I + \lambda_2 L^T L)x \;=\; A^T b.$$

See Golub, Hansen, and O'Leary (1999) for more details. Another regularization approach involves setting the small singular values of $[A \,|\, b]$ to zero. This is the *truncated TLS problem* discussed in Fierro, Golub, Hansen, and O'Leary (1997).

Problems

P6.3.1 Consider the TLS problem (6.3.2) with nonsingular D and T. (a) Show that if $\mathsf{rank}(A) < n$, then (6.3.2) has a solution if and only if $b \in \mathsf{ran}(A)$. (b) Show that if $\mathsf{rank}(A) = n$, then (6.3.2) has no